

SOLAR CELLS

Wafer-scale monolayer MoS₂ film integration for stable, efficient perovskite solar cells

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One of the primary challenges in commercializing perovskite solar cells (PSCs) is achieving both high power conversion efficiency (PCE) and sufficient stability. We integrate wafer-scale continuous monolayer MoS₂ buffers at the top and bottom of a perovskite layer through a transfer process. These films physically block ion migration of perovskite into carrier transport layers and chemically stabilize the formamidinium lead iodide phase through strong coordination interaction. Effective chemical passivation results from the formation of Pb-S bonds, and minority carriers are blocked through a type-I band alignment. Planar p-i-n PSCs (0.074 square centimeters) and modules (9.6 square centimeters) with MoS₂/perovskite/MoS₂ configuration achieve PCEs up to 26.2% (certified steady-state PCE of 25.9%) and 22.8%, respectively. Moreover, the devices show excellent damp heat (85°C and 85% relative humidity) stability with <5% PCE loss after 1200 hours and notable high temperature (85°C) operational stability with <4% PCE loss after 1200 hours.

Perovskite solar cells (PSCs) have attained power conversion efficiency (PCE) >26% (1, 2), but halide perovskites are prone to degradation during solar cell operations because of their soft lattice structures and relatively weak bonds. Even with encapsulation to isolate moisture and oxygen, the instability of perovskites under the stressors of heat, light irradiation, and electric field is still a key obstacle to be overcome before commercialization can be achieved (3). Specifically, heat and light easily activate ionic defects within the perovskite lattice and these defects mediate severe ion migration in perovskite films under electric field. Ion migration will lead to substantial changes in perovskite films followed by further degradation of other functional layers, which deteriorates the performance of PSCs under long-term operations (4). Therefore, reducing ion migration is essential to enhance the long-term stability of perovskite photovoltaics (PVs) (5).

Substantial research efforts have been devoted to suppression of ion migration. Interface engineering has become a feasible strategy to physically and/or chemically suppress ion migration by passivating mobile charged defects and eliminating the path of ion migration (6–10). Halide salts, inorganic oxysalts, organic mole-

cules, and polymers are the most frequently used passivators and form covalent bonds, ionic bonds, or other secondary interactions with perovskites (7, 11–15). Such chemical interactions will effectively passivate the negative- or positive-charged defects and substantially reduce the migration of cations and anions, particularly iodide ions. Moreover, various continuous barrier layers such as graphene derivatives and MXenes can be introduced into the perovskite upper interface or buried interface to block the ion migration path toward carrier transport layers (CTLs) (6, 16, 17). Passivators often take effect only locally in the interface and the barrier layer may also impede the transport of charge carriers to some extent because of poor charge conductivity or wide band gaps (18, 19). The alkylammonium halide deposited on a three-dimensional (3D) perovskite film has the potential to form a conformal and continuous layer but may suffer from structural dimensionality transformation under light or heat, which would lead to interface reconstruction and loss of initial properties (20, 21).

A stable interface that could adequately block ion migration but also not impede carrier transport could address these issues—ideally, a compact, ultrathin buffer layer on the perovskite surface. The deposition of inorganic substances such as metal oxides and metal fluorides can accommodate the formation of compact and ultrathin layers. However, the physical vapor deposition methods that are required damage the perovskite or fail to achieve chemical passivation (22). Alternatively, monolayer inorganic 2D materials are more promising candidates for interlayers given their (i) dangling free surfaces, which realize stable heterojunctions with perovskites by bonding (ii) atomic thickness, which minimizes carrier transport loss, and (iii)

continuous 2D morphology, which maximally blocks ion migration. In particular, semiconductor transition metal dichalcogenides (TMDCs) have high carrier mobilities, suitable band structures, and optical transparency (23–26). The TMDC few-layer flakes or nanosheets have been used as CTLs or interface layers in PSCs (27–33). However, it is still challenging to nondestructively fabricate a continuous and compact monolayer buffer layer to ultimately immobilize ions and minimize ion migration channels.

In this work, we integrated wafer-scale monolayer MoS₂ interlayers at both the hole transport layer (HTL)/perovskite and perovskite/electron transport layer (ETL) interfaces. Experimental characterizations and theoretical calculations demonstrated that the wafer-scale MoS₂ films physically blocked the ion migration of perovskite into CTLs, as well as chemically stabilized the α -phase FAPbI₃ through strong coordination interaction. The MoS₂ interlayers also provided effective chemical passivation through Pb-S bonds and minority-carrier blocking through type-I band alignment. The atomic thinness (<1 nm) of monolayer MoS₂ overcame the trade-off between passivation quality and series resistance and maximized both open circuit voltage (certified V_{OC} = 1.20 V) and fill factor (certified FF = 84.3%) of PSCs.

As a result, we achieved a certified steady-state PCE of 25.9% in inverted PSCs and a PCE of up to 22.8% in inverted perovskite mini-modules (9.6 cm²). Furthermore, resultant PSCs showed excellent damp-heat stability and operational stability. Notably, the inverted PSCs with MoS₂/FAPbI₃/MoS₂ configuration retained >95% of their initial PCE after 1200 hours at damp-heat test conditions [85°C and 85% relative humidity (RH)], and showed negligible efficiency decline after 2000 hours of maximum power point (MPP) tracking under continuous 1-sun illumination at room temperature (RT). The target device demonstrated notable high-temperature operational stability, maintaining >96% of its initial PCE after 1200 hours of MPP tracking at 85°C. Our study links halide perovskites and 2D materials through interface engineering, which not only expands the application of 2D materials but also provides a practical framework to improve the performance of perovskite-based optoelectronic devices.

Device fabrication

A wafer-scale uniform monolayer MoS₂ film was prepared on a soda-lime glass substrate through a face-to-face metal-precursor feeding route following our previous work (34). As shown in Fig. 1A, the color of glass substrate changed from transparent to light brown after the growth of MoS₂. Moreover, the typical optical microscopy of the sample surface showed highly uniform color contrast, indicating the homogenous thickness and in-plane continuity

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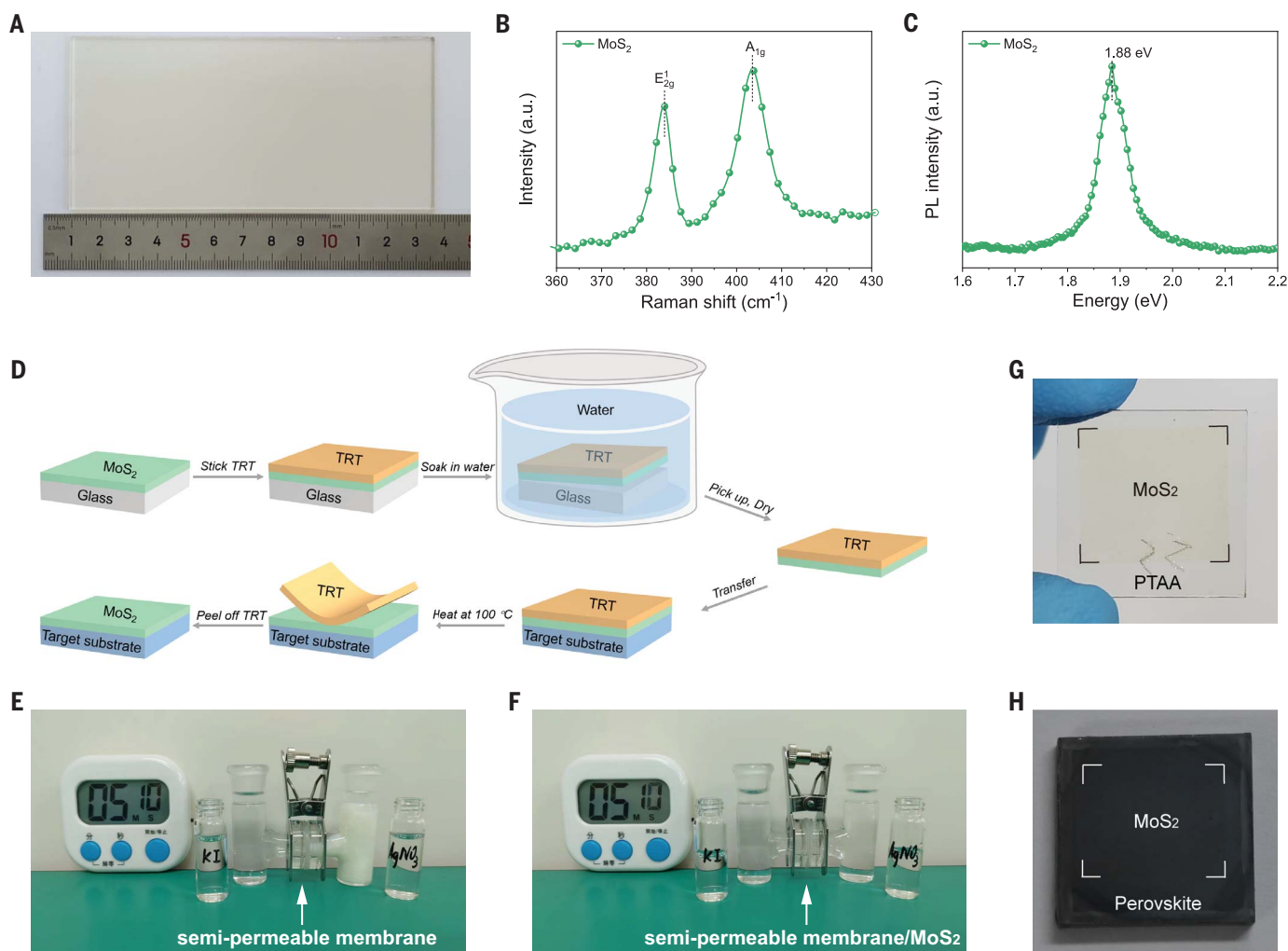


Fig. 1. Preparation, characterization, and transfer of monolayer MoS₂ films. (A) Photograph of the as-grown MoS₂ film on soda lime glass. (B) Raman and (C) PL spectra of the MoS₂ film. (D) Schematic diagram of the transfer process of the MoS₂ film. (E) and (F) Diffusion experiments in a H-type diaphragm electrolyzer, where the diaphragm is a semipermeable membrane without (E) and with MoS₂ (F). KI aqueous solution was added to the left side of the electrolyzer and after 5 min AgNO₃ aqueous solution was added to the right side of the electrolyzer. (G) and (H) Photographs of the MoS₂ films transferred onto the surfaces of PTAA (G) and perovskite (H).

of the MoS₂ film (fig. S1). A scanning electron microscopy micrograph showed that the domain size of the as-grown MoS₂ film reached hundreds of micrometers (fig. S2).

The thickness of the prepared MoS₂ film was determined by Raman spectroscopy analysis. As shown in Fig. 1B, the Raman spectrum of MoS₂ film showed two characteristic peaks located at 384.0 cm⁻¹ and 403.5 cm⁻¹, corresponding to the E_{2g}¹ and A_{1g} vibration modes, respectively. The difference of these two characteristic peaks was 19.5 cm⁻¹, indicating the monolayer nature of the obtained MoS₂ film (35). Moreover, the photoluminescence (PL) spectrum of the MoS₂ film presented a sharp and strong peak at ~1.88 eV (Fig. 1C), which was characteristic of typical MoS₂ A-exciton resonances (33).

To integrate the wafer-scale monolayer MoS₂ in perovskite devices, we developed a nondestructive and scalable transfer process (Fig. 1D). First, the edge of as-grown glass/MoS₂

was scratched with a knife to provide a path for water penetration. Then, thermal release tape (TRT) was stuck on the MoS₂ film and the glass/MoS₂/TRT was subsequently immersed in deionized water. The difference in surface energies of MoS₂ and the glass substrate caused water to penetrate into the interface. After MoS₂/TRT was completely delaminated from the glass substrate by the capillary force, it was picked up and then blowdried with N₂ flow. Next, the dried MoS₂/TRT was pressed onto a target substrate. Finally, the target substrate with MoS₂/TRT was heated to 100°C on a hotplate. The TRT lost adhesion at high temperature and automatically peeled off, leaving the MoS₂ film on the target substrate. The transferred MoS₂ film (fig. S3) showed a thickness of ~0.8 nm, which was the same as that of the previously reported value for a monolayer thickness (36).

We first explored the feasibility of MoS₂ physically blocking ion diffusion. We added KI

and AgNO₃ aqueous solutions to the left and right sides of an H-type diaphragm electrolyzer respectively, in a sequential manner. For the control sample (the diaphragm being a semipermeable membrane), I⁻ diffusion produced yellowish precipitates (Ag⁺ + I⁻ = AgI_s) after AgNO₃ was added (Fig. 1E). When the diaphragm was a semipermeable membrane attached with wafer-scale MoS₂, no visible precipitates were observed on either side of the electrolyzer after adding AgNO₃ (Fig. 1F), indicating that MoS₂ effectively blocked I⁻ diffusion. We expected that MoS₂ could also help block ion migration in PSCs.

On the basis of the above transfer process, wafer-scale MoS₂ films were integrated into the planar p-i-n PSCs where the perovskite composition was formamidinium lead iodide (FAPbI₃). The control devices used the architecture of indium tin oxide (ITO)/poly(triaryl)amine (PTAA)/perovskite/fullerene (C₆₀)/bathocuproine (BCP)/silver (Ag). Further, continuous monolayer MoS₂

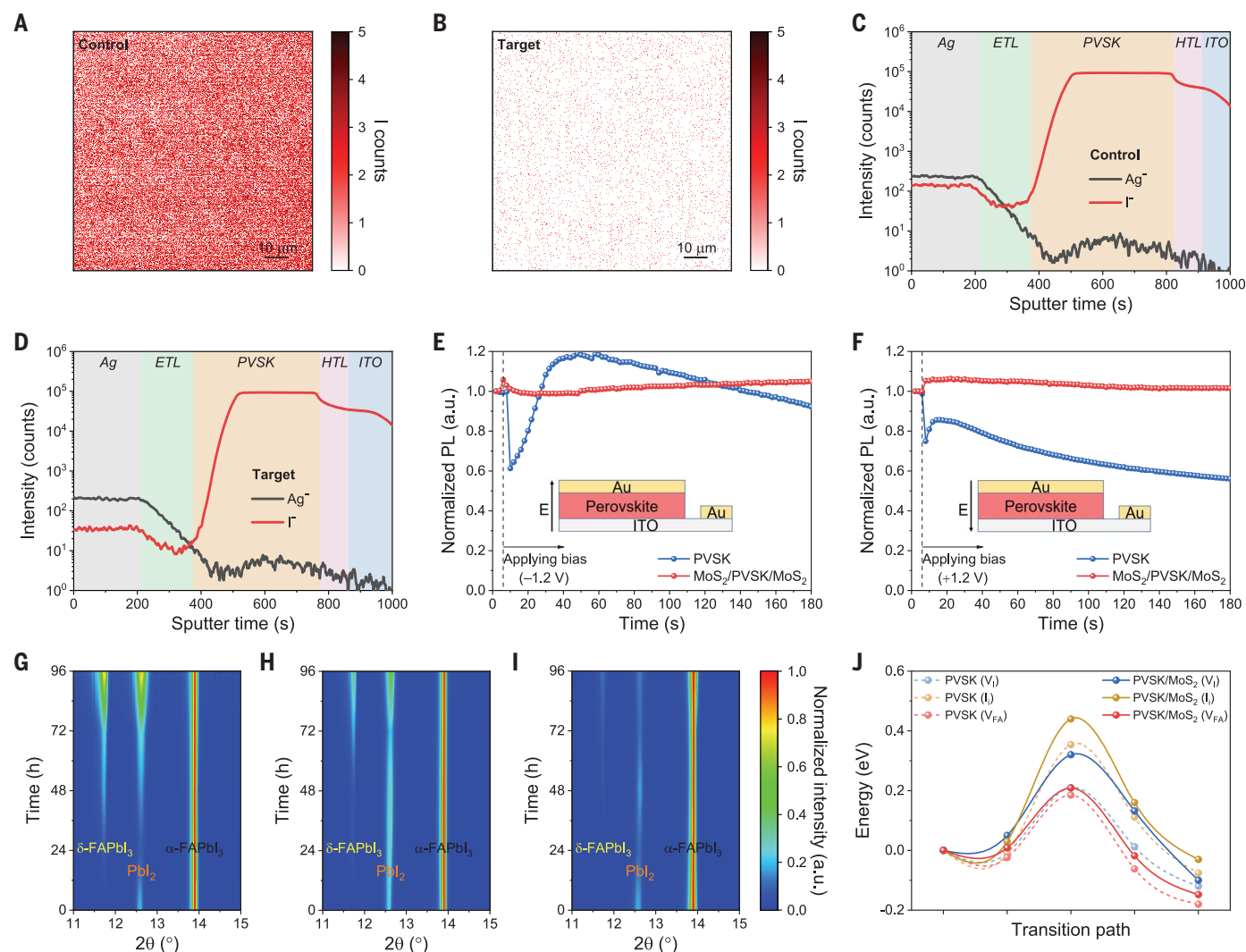


Fig. 2. Stabilization of configurations. (A) and (B) ToF-SIMS Mapping of I signals on Ag electrodes for ITO/FAPbI₃/Ag (A) and ITO/FAPbI₃/MoS₂/Ag (B) devices. The devices were aged at 85°C for 12 hours in a N₂ glove box. (C) and (D) Depth profiles of ToF-SIMS for aged control (C) and target (D) devices. The devices were aged at stabilized power output tracking under 1-sun illumination for 200 hours in a N₂

glove box. (E) and (F) Time-resolved PL intensity for ITO/FAPbI₃/Au or ITO/MoS₂/FAPbI₃/MoS₂/Au devices under -1.2 V (E) and +1.2 V (F) bias. (G to I) Evolution of XRD patterns of perovskite (G), MoS₂/perovskite (H), and MoS₂/perovskite/MoS₂ (I) films aged at 85% RH. (J) Energy curves of phase transition for the perovskite [(001) FAI-terminated FAPbI₃ surface with V_t, l_t, and V_{FA}] with and without MoS₂ passivation.

films were successively transferred to the surfaces of PTAA (Fig. 1G and fig. S4) and perovskite (Fig. 1H), resulting in target devices with MoS₂/perovskite/MoS₂ configuration. Like the perovskite film grown on PTAA (fig. S5A), the perovskite film grown on PTAA/MoS₂ still showed typical polycrystalline feature (fig. S5B) and α -phase structure (fig. S6). After the MoS₂ film was transferred to the upper surface of perovskite film, the morphology (fig. S5C) and structure (fig. S6) of the perovskite film were well-maintained, indicating that the transfer process did not appear to damage the perovskite.

Stabilized perovskite

We expected ion migration to be inhibited by the wafer-scale MoS₂ interlayer. We fabricated ITO/FAPbI₃/Ag and ITO/FAPbI₃/MoS₂/Ag de-

vices and aged them at 85°C for 12 hours in a nitrogen glove box. We then performed time-of-flight secondary ion mass spectroscopy (ToF-SIMS) measurements on these aged devices and analyzed the spatial distribution of I signals on Ag electrodes. Iodide ions, which are the most mobile ions in perovskite, easily escape from the perovskite and diffuse into the Ag electrode to form stable AgI compound (37). As a result, the Ag electrode of ITO/FAPbI₃/Ag sample was already fully filled with I (Fig. 2A). By contrast, only minimal I signal was observed in the Ag electrode of ITO/FAPbI₃/MoS₂/Ag sample, as shown in Fig. 2B. The results showed that the wafer-scale monolayer MoS₂ interlayer alleviated I⁻ migration into the Ag electrode.

We also aged the cells with the architecture of ITO/PTAA/FAPbI₃/C₆₀/BCP/Ag or ITO/PTAA/

MoS₂/FAPbI₃/MoS₂/C₆₀/BCP/Ag at stabilized power output tracking under 1-sun illumination for 200 hours in a nitrogen glove box (fig. S7). We then probed the depth profiles of atomic species in aged devices by ToF-SIMS. As shown in Fig. 2, C and D, the I signal was observed throughout the Ag electrode for the control cell, but the I signal in the Ag layer was substantially reduced for the target cell. The wafer-scale MoS₂ interlayer not only acted as an isolation layer to physically block the migration of perovskite ions to adjacent layers but also may have chemically passivated surface defects and eliminated the path of ion migration, as discussed below.

We further analyzed the PL evolution of the sample when a fixed bias was applied to the vertical ohmic devices ITO/FAPbI₃/Au or ITO/MoS₂/FAPbI₃/MoS₂/Au. To rule out the influence

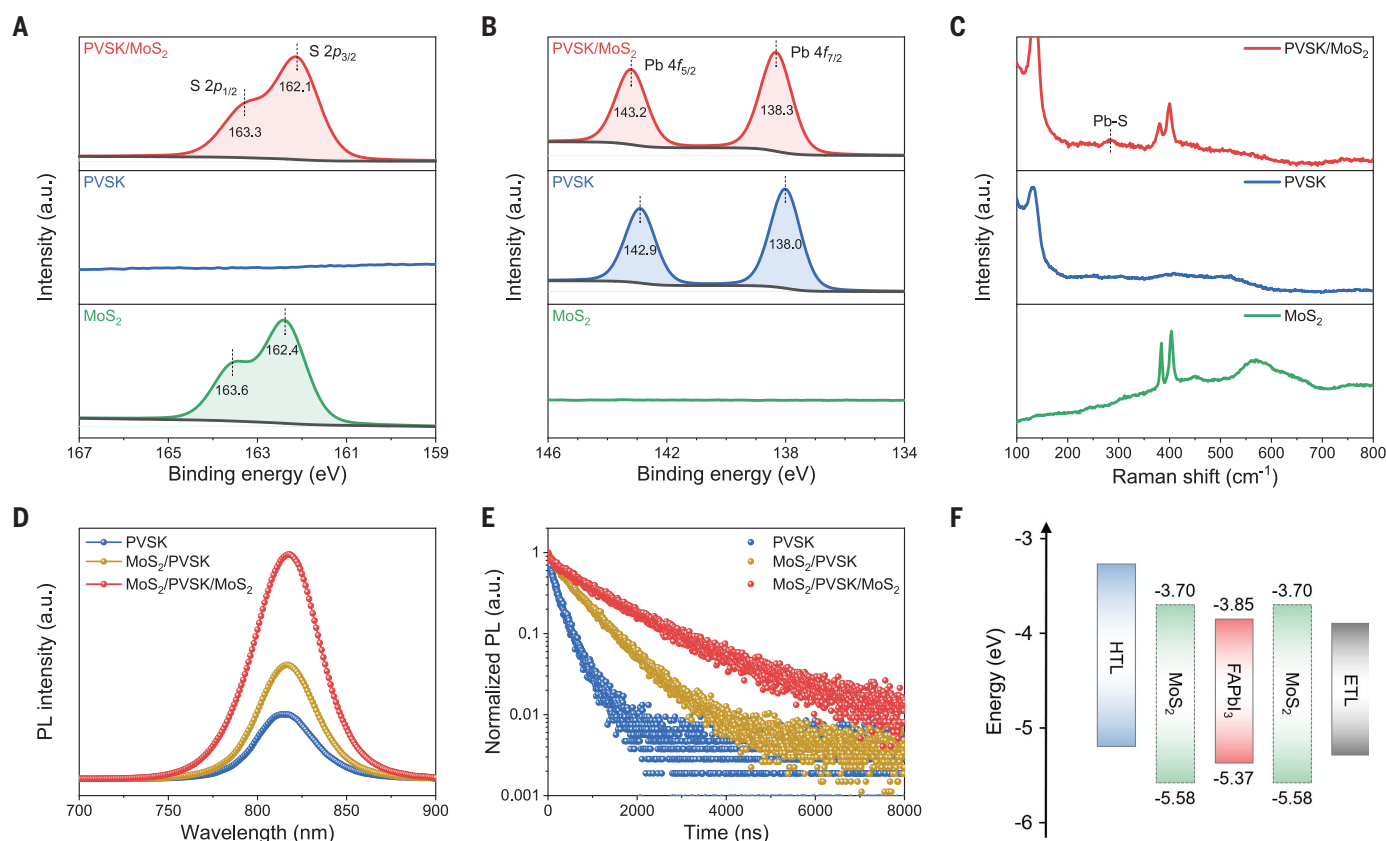


Fig. 3. Characterization of configurations. (A) and (B) XPS and (C) Raman spectra of MoS₂, perovskite, and perovskite/MoS₂ films. PL spectra (D) and TRPL decay curves (E) of perovskite, MoS₂/perovskite, and MoS₂/perovskite/MoS₂ films. (F) Energy-level alignment between perovskite and monolayer MoS₂.

of ambient atmosphere, each device was encapsulated with a glass slide and ultraviolet adhesive. The bias direction could be arbitrarily set as negative or positive, corresponding to the drift of iodine ions to the ITO/FAPbI₃ interface or FAPbI₃/Au interface, respectively. The voltage was set at 1.2 V, which is near the photovoltage of PSCs under 1-sun illumination. By applying a bias of -1.2 V (Fig. 2E) or $+1.2$ V (Fig. 2F), both the ITO/FAPbI₃/Au sample and ITO/MoS₂/FAPbI₃/MoS₂/Au sample showed instantaneous intensity fluctuation caused by the interference of electric field on the equilibrium carriers.

After a longer bias for 180 s, the PL intensity of ITO/FAPbI₃/Au sample decreased rapidly. We attributed the decay of the PL intensity to electric field-driven ion migration, whereby the perovskite film accumulates a high density of defects (38, 39). By contrast, regardless of the applied bias direction (-1.2 V or $+1.2$ V), ITO/MoS₂/FAPbI₃/MoS₂/Au sample maintained a stable PL intensity throughout the entire measurement period. These observations have rarely been achieved, which indicate that ion migration in the perovskite film was greatly suppressed by the wafer-scale MoS₂ interlayers.

We also found that the MoS₂ interlayer helped stabilize the α -phase FAPbI₃. The photoactive

FAPbI₃ (α -phase) is metastable and undergoes phase transition to the non-photoactive δ -phase in an ambient humid atmosphere (40). As shown in Fig. 2, G to I, we performed accelerated aging of unencapsulated samples (pristine perovskite, MoS₂/FAPbI₃ and MoS₂/FAPbI₃/MoS₂), at 85% RH and tracked the evolution of the crystalline phases in the aged films by x-ray diffraction (XRD) measurements. By directly exposing the pristine perovskite film (Fig. 2G) to high humidity (85% RH), a diffraction peak at $2\theta = 11.7^\circ$ assigned to the (010) plane of δ -FAPbI₃ appeared after six hours. By contrast, the appearance of δ -FAPbI₃ in the MoS₂/FAPbI₃ film (Fig. 2H) was delayed to 12 hours.

Moreover, after 96 hours of aging, the relative intensity of the diffraction peak at $2\theta = 11.7^\circ$ in the MoS₂/FAPbI₃ film was also lower than that in the pristine perovskite film. In this case, it may be related to the interaction between MoS₂ at the bottom and perovskite that inhibits the phase transition. The phase transition from black perovskite to yellow non-perovskite deforms the [PbI₆] octahedrons from corner-sharing to face-sharing, and we infer that the interaction at the interface restricts this structural transformation (6, 8).

When the MoS₂/FAPbI₃/MoS₂ configuration was tested, only a slight δ -phase appeared dur-

ing the entire 96-hour aging process (Fig. 2I). In addition to the interaction, MoS₂ protecting the perovskite from moisture invasion also inhibited the phase transition and contributed to the humidity stability. We aged unencapsulated cells with different architectures at room temperature (RT) and 85% RH. After 200 hours of aging, ITO/PTAA/MoS₂/FAPbI₃/MoS₂/C₆₀/BCP/Ag devices retained $\sim 79\%$ of initial PCE, compared with the retention of ~ 14 and $\sim 33\%$ for ITO/PTAA/FAPbI₃/C₆₀/BCP/Ag and ITO/PTAA/MoS₂/FAPbI₃/C₆₀/BCP/Ag devices, respectively (fig. S8).

We performed first-principles density functional theory (DFT) simulation to investigate the phase transition. We focused on the (001) FAI-terminated FAPbI₃ surface with three typical point defects [I vacancies (V_I), I interstitials (I_i) and FA vacancies (V_{FA})] in touch with monolayer MoS₂, as shown in fig. S9. The energy barrier of phase transition from α -phase to δ -phase was calculated to be 0.210, 0.354, and 0.185 eV for pristine perovskite with V_I , I_i , and V_{FA} , respectively, which increased to 0.320, 0.440, and 0.209 eV, respectively, after the introduction of MoS₂ (Fig. 2J). We also considered a PbI₂-terminated FAPbI₃ surface with V_I , I_i , and Pb vacancies (fig. S10A), and simulation results showed that the energy barrier of phase

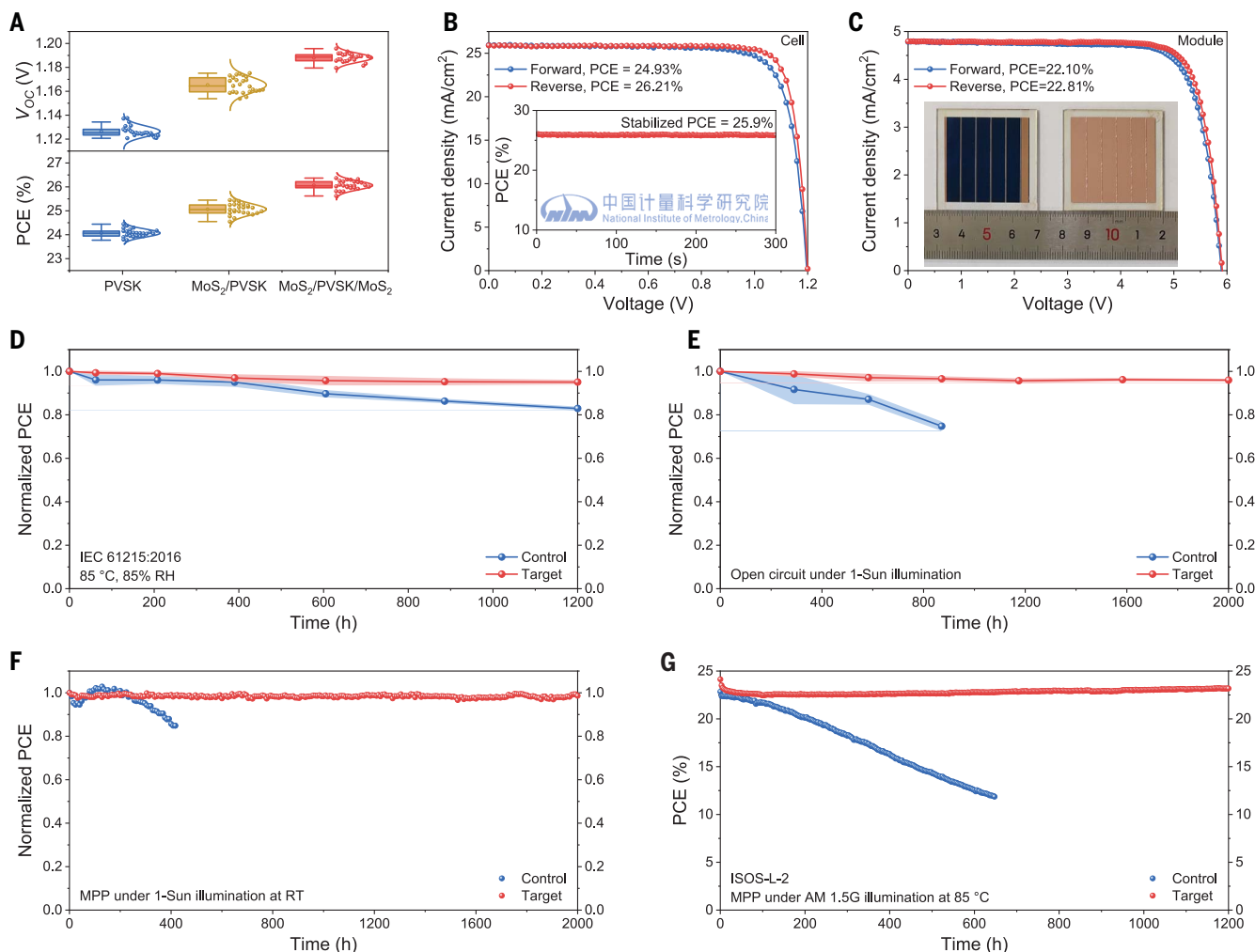


Fig. 4. Performance of PSCs. (A) PCE and V_{OC} statistics of solar cells with different configurations. (B) $J-V$ curves and stabilized power output (inset) of the best-performing target cell measured at National Institute of Metrology, China. (C) $J-V$ curves and photograph (inset) of the best-performing minimodule with target architecture. (D) Damp-heat tests for encapsulated

devices at 85°C and 85% RH. (E) Photostability of devices at open circuit under 1-sun illumination in a nitrogen atmosphere. (F) MPP tracking for devices under 1-sun illumination at room temperature in a nitrogen atmosphere. (G) MPP tracking for encapsulated devices under AM 1.5G illumination at 85°C in the ambient atmosphere.

transition increased from 0.171, 0.425, and 0.153 eV to 0.196, 0.488, and 0.286 eV, respectively, after MoS_2 passivation (fig. S10B). These simulation results were consistent with the experimental results, which demonstrated that the interaction between MoS_2 and FAPbI_3 inhibited the phase transition of $\alpha\text{-FAPbI}_3$ to $\delta\text{-FAPbI}_3$.

Interface passivation

To explore the interaction between the MoS_2 interlayer and perovskite absorber, we used x-ray photoelectron spectroscopy (XPS). In the perovskite/ MoS_2 film, there were additional S 2p peaks (Fig. 3A), again indicating the presence of MoS_2 on the perovskite surface. The S 2p core level of perovskite/ MoS_2 shifted to a lower binding energy than that of the pristine MoS_2 , and the Pb 4f core level (Fig. 3B) of perovskite/

MoS_2 shifted to a higher binding energy than that of the pristine perovskite. These energy level shifts denoted changes in chemical states that could be attributed to the strong bonding between MoS_2 and perovskite (9, 10).

We performed the Raman spectroscopy characterization to further characterize the bonding between perovskite and MoS_2 . As shown in Fig. 3C, the Raman spectrum of the perovskite/ MoS_2 film showed an additional peak located at $\sim 284\text{ cm}^{-1}$, which we assigned to a Pb-S stretch (41). The Raman result was in good agreement with the XPS result, indicating the formation of Pb-S bonds between the perovskite surface and monolayer MoS_2 . As a result, strong Pb-S bonding was beneficial to chemically stabilize perovskite.

Furthermore, the Pb-S bonds chemically passivated undercoordinated Pb at perovskite in-

terfaces, which effectively reduced nonradiative recombination losses. We prepared pristine perovskite, $\text{MoS}_2/\text{perovskite}$, and $\text{MoS}_2/\text{perovskite}/\text{MoS}_2$ samples and performed PL and time-resolved PL (TRPL) measurements. The result (Fig. 3D) showed that the $\text{MoS}_2/\text{perovskite}$ sample exhibited 1.8 times higher PL intensity than pristine perovskite film. It was consistent with the TRPL measurement (Fig. 3E), wherein the $\text{MoS}_2/\text{perovskite}$ sample showed a longer carrier lifetime ($\tau_1 = 164.9\text{ ns}$ and $\tau_2 = 727.8\text{ ns}$) than the pristine perovskite film ($\tau_1 = 90.0\text{ ns}$ and $\tau_2 = 336.3\text{ ns}$). When $\text{MoS}_2/\text{perovskite}/\text{MoS}_2$ was integrated, the steady-state PL intensity was further enhanced to 3.5 times that of the pristine perovskite film, and the PL decay lifetime was substantially extended to 214.3 ns (τ_1) and 1384.8 ns (τ_2). Nonradiative recombination of photo-generated carriers was largely

reduced by interface passivation by the wafer-scale MoS₂ layers.

We also obtained the energy levels of perovskite (fig. S11) and monolayer MoS₂ (fig. S12) films by ultraviolet photoelectron spectroscopy and ultraviolet-visible absorption spectra. According to band-edge analysis, the valance band maximum of perovskite (−5.37 eV) was higher than that of the monolayer MoS₂ (−5.58 eV) whereas the conduction band minimum of perovskite (−3.85 eV) was lower than that of monolayer MoS₂ (−3.70 eV). Notably, perovskite and MoS₂ contact formed a typical type-I band alignment (Fig. 3F), in which majority carriers tunneled and minority carriers returned (42). Thus, MoS₂ provided a spatial separation and energy barrier to reduce recombination between the holes (or electrons) in the perovskite layer and the electrons in the ETL (or holes in the HTL) (43).

Transient photovoltage and transient photocurrent measurements were employed to investigate the carrier dynamics in cells with different configurations. As shown in fig. S13, devices with MoS₂ interlayers exhibited longer carrier recombination lifetimes, indicating reduced nonradiative recombination associated with corresponding interfaces. Notably, devices with MoS₂ interlayers did not exhibit slowed carrier transport, suggesting that majority carriers tunnel smoothly through monolayer MoS₂.

Improved device performance

We investigated the effect of MoS₂ interlayers on the PV performance of the completed cells. The detailed PV parameters of cells with different configurations were determined from the current-voltage (*J-V*) sweep (fig. S14). MoS₂ interlayer at the HTL/perovskite interface improved the mean PCE of the cells from 24.1 to 25.1%, and the mean PCE of the cells further increased to 26.1% when the MoS₂ interlayer at the perovskite/ETL interface was further applied (Fig. 4A and table S1). The statistical results showed that the mean *V*_{OC} increased from 1.13 V to 1.19 V in average. The improved *V*_{OC} was attributed to the MoS₂ interlayers, which passivated perovskite interfaces to reduce nonradiative recombination. Moreover, the similar short-circuit current density (*J*_{SC}) and slightly increased FF (figs. S15 and S16) indicated that monolayer MoS₂ enabled unimpeded carrier transport.

The best-performing device was sent to an accredited PV laboratory (the PV Metrology Lab of National Institute of Metrology, China) for certification, and the test results shown in Figs. 4B, S17, and S18. The best-performing cell with the architecture of ITO/PTAA/MoS₂/FAPbI₃/MoS₂/C₆₀/BCP/Ag achieved a certified PCE of 25.6%, which was the average value of the forward and reverse sweeps. By holding a bias at the maximum power output point (1.05 V),

we obtained a certified efficiency of ~25.9% from the steady-state measurement.

We tested the *J-V* response of PSCs under different light intensities (fig. S19, A to C). We found that all three devices showed a linear relation between *J*_{SC} and light intensity (fig. S19D), indicating good charge transport when MoS₂ buffers were introduced. Figure S19E showed a linear relation between *V*_{OC} and the logarithm of light intensity with a slope of $nk_B T/q$, where *n* is the ideality factor, *k_B* is the Boltzmann constant, *T* is the temperature, and *q* is the electron charge. The control cell showed a slope of $1.96k_B T/q$ whereas the target cell showed a much smaller slope of $1.35k_B T/q$. The decrease in *n* further supported our finding that MoS₂ interlayers greatly reduced trap-assisted recombination.

The universality of the monolayer MoS₂ buffer was demonstrated by applying it to another perovskite component (FA_{1-x}Cs_xPbI₃) and n-i-p device structure. As shown in figs. S20 and S21, we observed improved device performance for all cases upon MoS₂ introduction. We also fabricated perovskite minimodules with aperture areas of 9.6 cm². As shown in Fig. 4C, the minimodule based on the MoS₂/perovskite/MoS₂ configuration with five subcells showed high aperture efficiency of 22.8% for the reverse sweep (with *V*_{OC} of 5.91 V, *J*_{SC} of 4.79 mA/cm², and FF of 80.5%) and 22.1% for the forward sweep (with *V*_{OC} of 5.89 V, *J*_{SC} of 4.78 mA/cm², and FF of 78.4%). The corresponding active-area efficiency was 24.1% (reverse sweep) and 23.3% (forward sweep) with a geometric fill factor of 94.8%. However, the highest aperture efficiency of the control minimodules was only 20.4% (active-area efficiency of 21.5%) derived from *J-V* scanning (fig. S22). These results confirmed that the MoS₂ buffer was scalable for achieving high-performance large-sized modules.

To further enhance the device stability, 10 nm SnO₂ and 100 nm ITO were deposited on top of the C₆₀ sequentially to replace the BCP layer (figs. S23 and S24). Subsequently, we performed a series of stability investigations on corresponding devices. First, we encapsulated each PSC in two glass sheets with polyolefin elastomer encapsulant and edge sealing with butyl rubber (fig. S25). Then, the encapsulated cells were subjected to damp-heat tests (85°C and 85% RH, International Electrotechnical Commission (IEC) 61215:2016 protocol) in a damp-heating chamber (fig. S26). As shown in Fig. 4D, the target devices with MoS₂/perovskite/MoS₂ retained >95% of their initial PCE after 1200 hours whereas control devices retained 82.9% of their initial PCE.

We also investigated the photostability of devices at open circuit under continuous 1-sun illumination. As shown in Fig. 4E, the target devices maintained 96.6% of their initial PCE after 2000 hours, which was superior to that of the control. We examined the operational stability of devices under continuous 1-sun illumination at room temperature by tracking their MPPTs.

As shown in Fig. 4F, the target device showed negligible efficiency decline after 2000 hours of MPP tracking, whereas the control device presented a PCE loss of ~15% after only 400 hours. We further investigated the high-temperature operational stability by MPP tracking for encapsulated devices under AM 1.5G illumination at 85°C in the ambient atmosphere (International Summit on Organic Photovoltaic Stability (ISOS)-Light soaking tests (L)-2 protocol). After 1200 hours of continuous operation, the target device maintained an efficiency of more than 23% (>96% of original PCE). A drop in initial efficiency in the first several hours was likely caused by ion migration inside the perovskite (5, 44). By contrast, the control device was reduced to <55% of initial PCE after only 600 hours (Fig. 4G). More stability tests are shown in figs. S27 to S30 in the SM.

The above experiments confirmed that MoS₂ interlayers significantly improved the long-term stability of PSCs, which was attributed to the stable MoS₂/perovskite/MoS₂ configuration with strong bonding. Wafer-scale MoS₂ buffers, which served as moisture/oxygen ingress barriers and ion migration blocks, stabilized the structure and components of perovskites, thus enhancing the resistance of perovskites against the stressors of moisture, heat, light, and electric field. Further, we extended the interlayer to other TMDCs such as monolayer WS₂. As shown in fig. S31, the statistical results showed that the device PCE increased significantly from 23.4 to 25.3% on average when WS₂/perovskite/WS₂ was applied, close to that of MoS₂/perovskite/MoS₂ (25.5%). Moreover, the device with WS₂/perovskite/WS₂ configuration showed impressive operational stability with minimal PCE loss after 300 hours of MPP tracking, comparable with the device with MoS₂/perovskite/MoS₂ configuration (fig. S32). These results indicated that monolayer WS₂ films also acted as ideal buffers at perovskite interfaces, demonstrating the universality of our strategy.

Discussion

We have demonstrated that wafer-scale monolayer MoS₂ films can act as ideal buffers at the top and bottom of a perovskite layer. The continuous wafer-scale monolayer 2D morphology can physically block ion migration and ultimately immobilize ions and minimize ion migration channels, thus achieving excellent stability of PSCs. Meanwhile, the strong bonding between perovskite and MoS₂ chemically passivates the defects of perovskite, thereby reducing the non-radiative recombination of carriers at perovskite interfaces to improve the device efficiency.

To date, the development of wafer-scale monolayer inorganic 2D films to minimize the ion migration in the interface have not been reported in perovskite-based devices. It leverages the excellent properties of 2D materials (e.g., dangling free surface, atomic thickness,

and continuous 2D morphology) to make efficient and stable halide PSCs. Moreover, the findings bridge the 2D materials and soft lattice optoelectronic materials for heterojunction engineering, which can extend to the construction of highly efficient and stable devices in other related fields.

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ACKNOWLEDGMENTS

Funding: This work was supported by the National Natural Science Foundation of China (52125206, 52302320, 52433013, and 51925201), the Beijing Natural Science Foundation (Z240024), the National Key Research and Development Program of China (2023YFB4202502 and 2020YFB1506400), the Fundamental

Research Funds for the Central Universities (2652022008), the Tencent Foundation through the Xplorer Prize, the China National Petroleum Corporation-Peking University Strategic Cooperation Project of Fundamental Research, the Sinopec Seeding Program, and the Yunnan Provincial Science and Technology Project at Southwest United Graduate School (202302A0370013). **Author contributions:** H. Zai and H. Zhou conceived and supervised the project. H. Zai fabricated and characterized the films and devices and performed the stability test. P.Y., T.Z., and Ya. Zhang prepared monolayer MoS₂ and WS₂ films. J.S. and J.C. performed the DFT calculations. R.Y., X.Z., Y.M., and C.Z. assisted in improving the efficiency of PSCs. R.F. performed the PL and TRPL measurements. Y.W. encapsulated the devices and performed the stability tests. W.Z. performed the XRD measurements. Yu Zhang measured PL evolution with bias. Zi. H., Y.J., N.L., Zh. H., and Q.C. advised on the research. Y.B. performed the ToF-SIMS measurements. H. Zai wrote the first draft of the manuscript. H. Zai and H. Zhou revised the manuscript. All authors contributed to the discussion and commented on the manuscript. **Competing interests:** Authors declare that they have no competing interests. **Data and materials availability:** All data are available in the main text or SM. **License information:** Copyright © 2025 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original US government works. <https://www.science.org/content/page/science-licenses-journal-article-reuse>

SUPPLEMENTARY MATERIALS

science.org/doi/10.1126/science.ado2351
Materials and Methods
Figs. S1 to S32
Tables S1 and S2
Movie S1

Submitted 25 January 2024; resubmitted 3 June 2024
Accepted 27 November 2024
[10.1126/science.ado2351](https://doi.org/10.1126/science.ado2351)